

| Sl.no | Model                  | Parameter 2                | Parameter 2          | R-score                   |
|-------|------------------------|----------------------------|----------------------|---------------------------|
| 1     | Muliple linear         | NA                         | NA                   | 0.789479035               |
| 2     | Support vector Machine | Kernel = linear            | c=0.1                | -0.122076684              |
|       |                        |                            | c=0.01               | -0.079762068              |
|       |                        | Kernel = polyomial         | c=0.1                | -0.08625251710262294      |
|       |                        |                            | C=0.01               | -0.08930331146175452      |
|       |                        | Kernel = rbf               | C=0.1                | -0.08957624598812952      |
|       |                        |                            | C=0.01               | -0.08969568284239293      |
|       |                        | Kernel ='sigmoid'          | C=0.1                | -0.08974351910465961      |
|       |                        |                            | C=0.01               | -0.08971245575310527      |
|       |                        |                            | C= 10                | -0.09078319814614         |
| 3     | Decision tree          | Squarer error              | NA                   | 0.7044466071937253        |
|       |                        | Criterion = squared error  | Splitter = best      | 0.6957219144244762        |
|       |                        |                            | Splitter = random    | 0.7210197393645568        |
|       |                        | Criterion = absolute error | Splitter = best      | 0.6680727002972597        |
|       |                        |                            | Splitter = random    | 0.7506349846089113        |
|       |                        | Criterion = poisson        | Splitter = best      | 0.7288161987230048        |
|       |                        |                            | Splitter = random    | 0.7019494699875246        |
| 4     | <b>Random Forest</b>   | n-estimator = 100          | Poisson              | 0.8526334258892607        |
|       |                        | <b>n-estimator = 100</b>   | <b>Squared error</b> | <b>0.8538307913484513</b> |
|       |                        | n-estimator = 100          | Absolute error       | 0.8520093621081837        |

Out of 4 Algorithm, Random Forest gives higher value for the arguments (n-estimator = 100,criterion = squared\_error) **0.8538307913484513**. Hence random forest is considered as the best model